Quick Sort

Definition

Quick Sort is a **divide-and-conquer sorting algorithm**. It selects a **pivot**, places smaller elements before it and larger elements after it, and then recursively sorts the two parts.

Algorithm

1. Take an array as input.
2. Select an element as the **pivot**.
3. Divide the elements into:
    a. Elements smaller than pivot
    b. Elements equal to pivot
    c. Elements greater than pivot
4. Recursively apply Quick Sort to the left and right parts.
5. Combine the sorted parts.
6. Display the sorted array.

Output:

```
Enter number of elements: 5
Enter element 1: 4
Enter element 2: 8
Enter element 3: 9
Enter element 4: 3
Enter element 5: 6
Original array: [4, 8, 9, 3, 6]
Sorted array: [3, 4, 6, 8, 9]
Execution time: 0.00013406999642029405 seconds


...Program finished with exit code 0
Press ENTER to exit console.
```

Space Complexity

For the **Python code I gave above**:

Space Complexity = O(n)

Because the code creates separate `left` and `right` lists during partitioning and uses recursion.

Short Answer for Viva

Time Complexity:

- Best: **O(n log n)**
- Average: **O(n log n)**
- Worst: **O(n^2)**

Space Complexity: O(n) for this Python implementation.